

Machine Learning Principles

Class17 : November 6

Bayesian Net I

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Today's Lecture

1. Bayesian net is a DAG representation of a joint density
 - *conditional independence
 - *posterior prob
2. Elementary blocks for Bayesian network (three nodes and two edges)
 - chain structure
 - tent (fork) structure
 - v-structure
3. D-separation:
determining a conditional independency of R.Vs given a Bayesian net.
4. Inference: querying posterior probability
 - (1) Variable Elimination
 - (2) Markov Chain Monte Carlo (MCMC) simulation

[1] Bayesian Net (generative modeling)

- Bayesian net learns a joint density.
- $P(X, Y)$
- Generative modeling
- Discriminative modeling
: aims to learn $P(Y|X)$

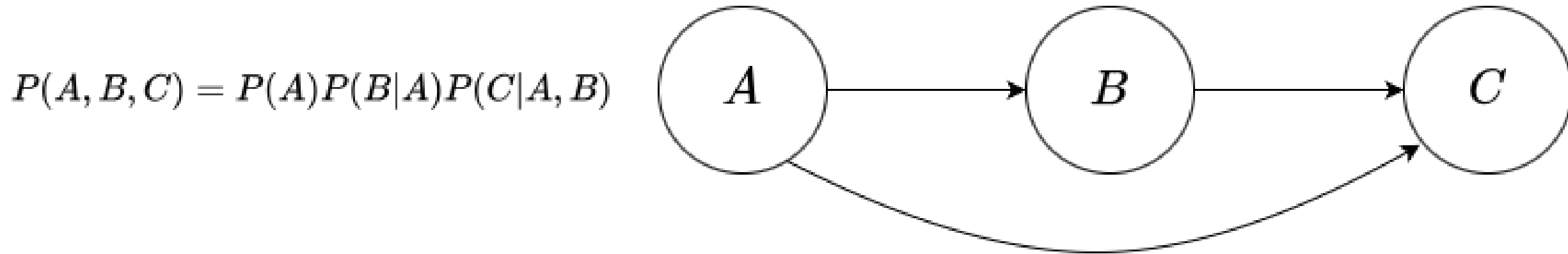
Q: which one is more flexible in terms of what task it can handle?

[2] Bayesian Net (generative modeling)

Given a generative modeling $P(X, Y)$,
we can query for the information $P(X|Y)$ or $P(Y|X)$.

[3] Bayesian Net (learning connectivity & CPT: conditional probability)

- A joint density
can be represented by the product of conditional factors
by a **Directed Acyclic Graph (DAG)**: Bayesian Net



Q: learning a Bayesian Net: learning what?

(1) connectivity (edges) (2) conditional densities for each vertex

[4] Bayesian Net (learning connectivity & CPT: conditional probability)

Q: graphs?

$$P(A, B, C) = P(A)P(B|A)P(C|B)$$

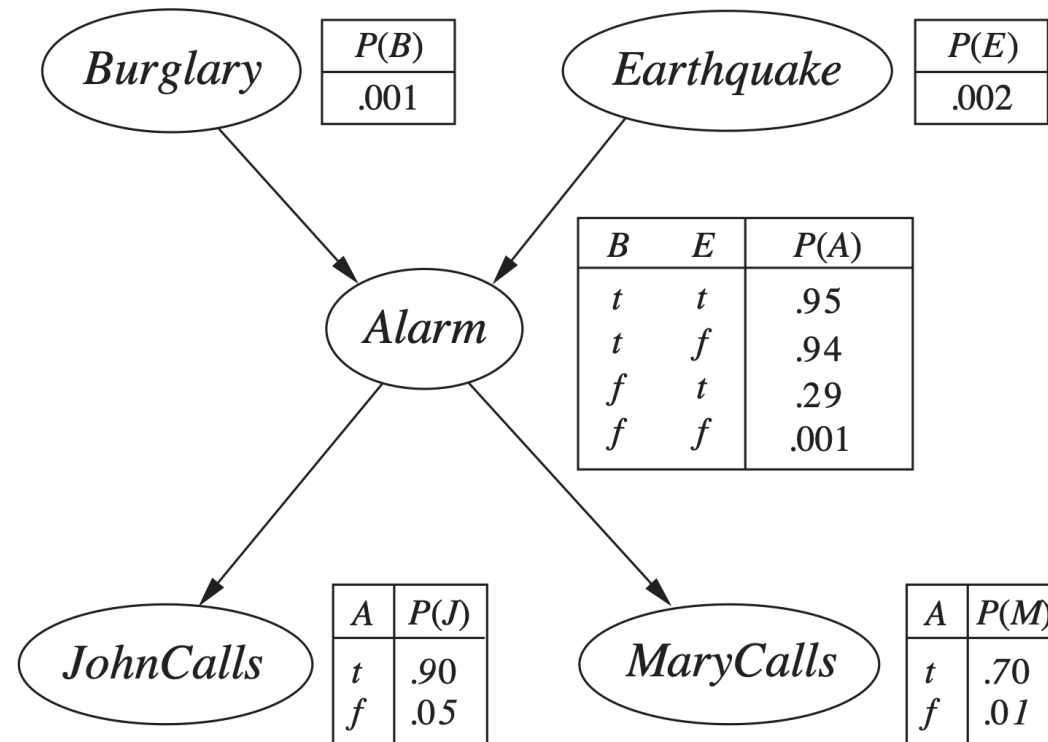
[5] Bayesian Net (learning connectivity & CPT: conditional probability)

Q: graphs?

$$P(A, B, C) = P(A)P(B)P(C)$$

[6] Bayesian Net (example)

Figure 14.2 From the book “AI: A Modern Approach”



- structure: a DAG encodes conditional independence relations among R.Vs.
- parameters: the conditional densities. (CPT: Conditional Probability Table)

[7] Bayesian Net (what we can do with BN?**)

[1] determine if a set of random variables are conditionally independent given some observation?

sometimes it is not decidable.

[2] cause and effect $P(B|C,A) = P(B|C) \rightarrow A$ is not the cause of B

[3] querying the posterior prob like $P(X,Y|z)$?

[8] Bayesian Net (causal and effect: **why** an event happens?)

- fire is the cause of setting off a fire alarm?
- high temperature is a cause of ice cream melting?

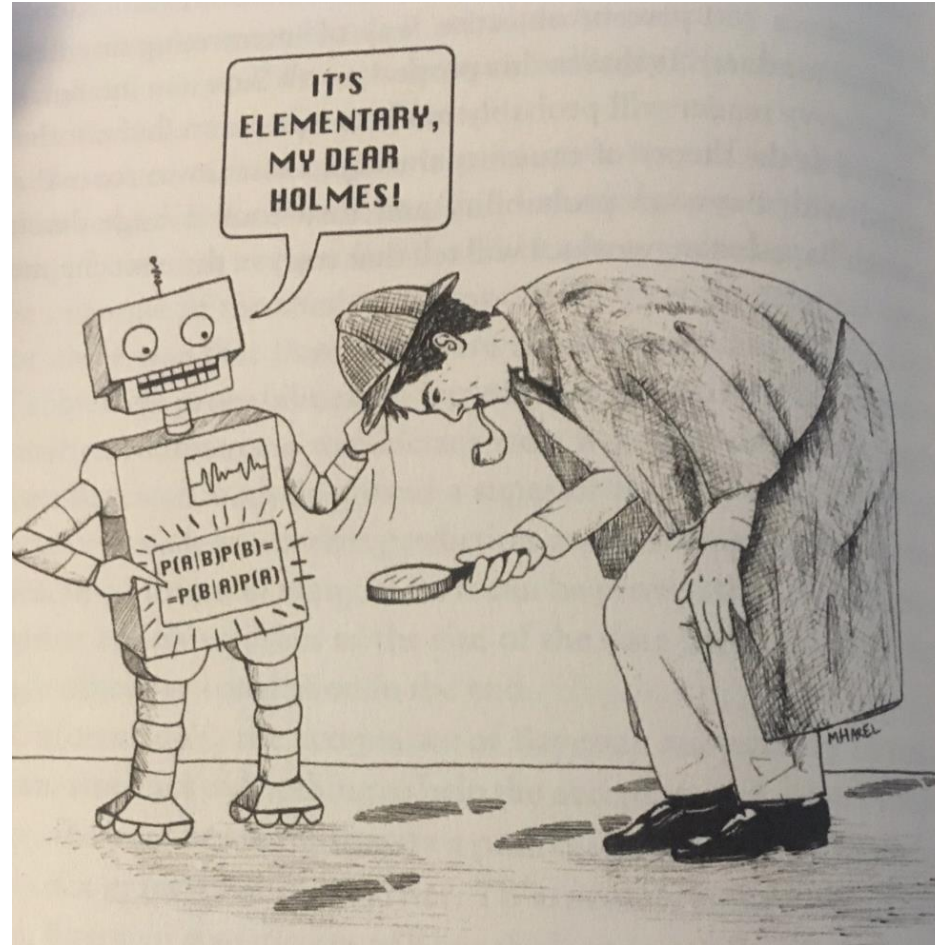
what evidence will be required to say A is a cause of B ? is it possible to say so?
or what evidence will be required to say A is not a cause of B ?

[9] Bayesian Net

The questions on uncertainty in our daily life?

- (1) when a fire alarm rings (observation),
what is the **chance of the fire has actually happened?**
- (2) what was the **content of the original letter (observation)**
given only a fragment of the paper?
- (3) why do we feel better after seeing a doctor? What is **the actual cause of our recovery? (the medicine I took last night or the natural progression of time?)**

[10] Bayesian Net



- Bayesian net provides a probabilistic tool help us systematically the answer these questions.

- Bayesian Net (some features)
:Joint Density Representation

[1] Bayesian Net Representation (different orders but the same density)

$$P(A, B, C) = P(A)P(B|A)P(C|A, B)$$

$$P(A, B, C) = P(C)P(B|C)P(A|B, C)$$

$$P(A, B, C) = P(B)P(A|B)P(C|A, B)$$

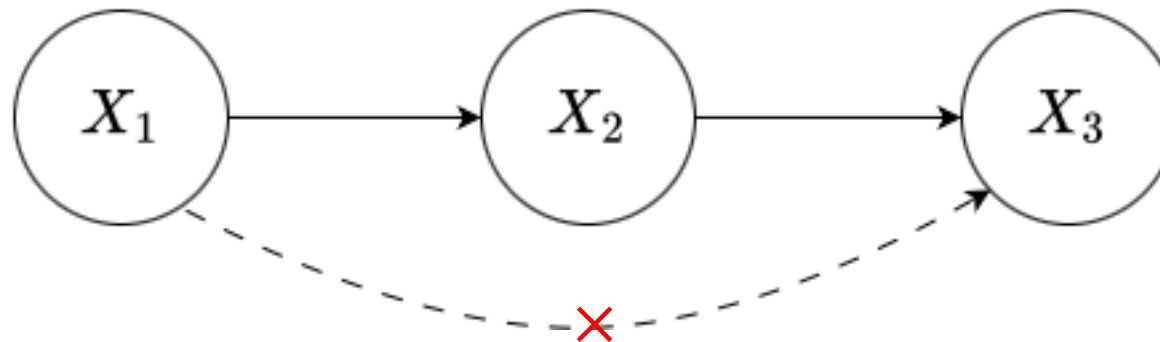
- A joint density can be represented in different orders with different CPTs.

[2] Bayesian Net Representation (different orders but the same density)

A joint density can be represented in various Bayes nets.
(depending on orders), but the Bayes nets encode the same joint density.

[3] Bayesian Net Representation (missing edges encodes CI)

$$P(X_1) P(X_2 | X_1) P(X_3 | X_2)$$



[4] Bayesian Net Representation (missing edges encodes CI)

- $P(X_1) P(X_2|X_1) P(X_3|X_2)$
- $[X_3 \perp X_1 \mid X_2]$
: X_3 and X_1 are conditionally independent given X_2 .

$$\begin{aligned} P(X_1, X_3|X_2) &= \frac{P(X_1, X_2, X_3)}{P(X_2)} \\ &= \frac{P(X_1, X_2, X_3)}{P(X_2)} \\ &= \frac{P(X_1)P(X_2|X_1)P(X_3|X_2)}{P(X_2)} \\ &= \frac{P(X_1, X_2)P(X_3|X_2)}{P(X_2)} = P(X_1|X_2)P(X_3|X_2) \end{aligned}$$

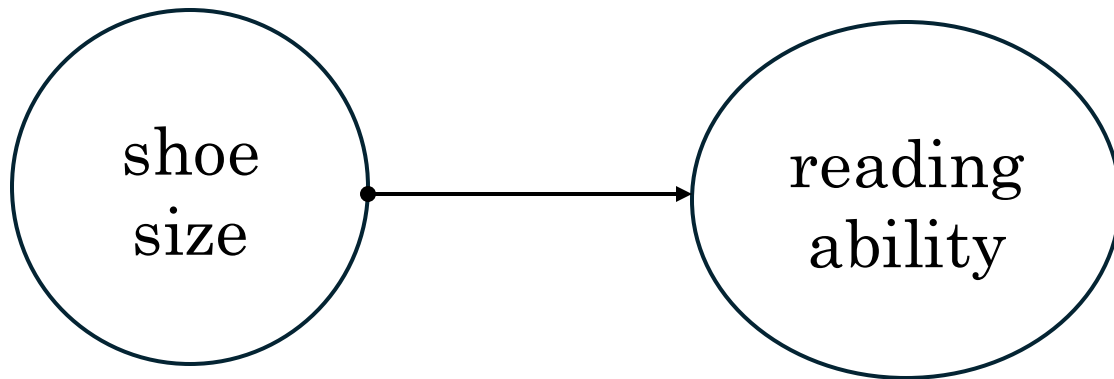
- In a Bayesian network,
the lack of edges indicates conditional independence in the forward direction.

[5] Bayesian Net Representation (causal and effect?)

Q: arrows A to B in a Bayesian network implies a causal (A) and effect (B) relationship?

[6] Bayesian Net Representation (causal and effect?)

Q: arrows A to B in a Bayesian network implies
a causal (A) and effect (B) relationship? they encode correlation and
not cause & effect

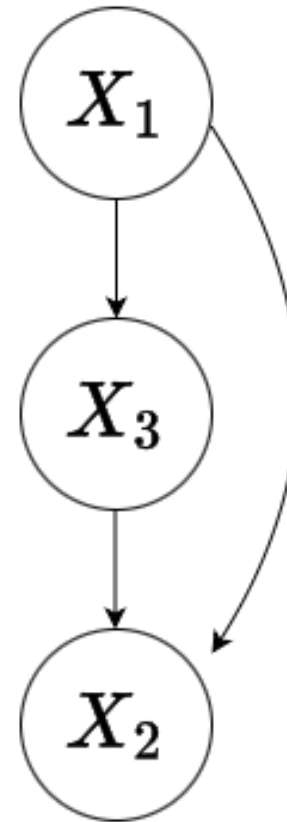
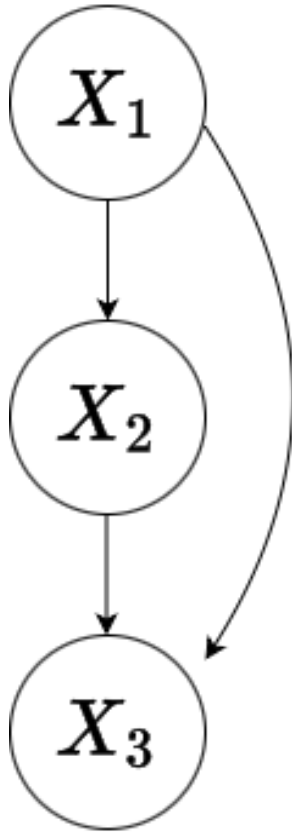


[7] Bayesian Net Representation (causal and effect?)

A joint density can be represented by several Bayes Nets.

There will be a contradiction if the arrows necessarily imply causation.

However, sometimes they can encode the relation causal and effect.

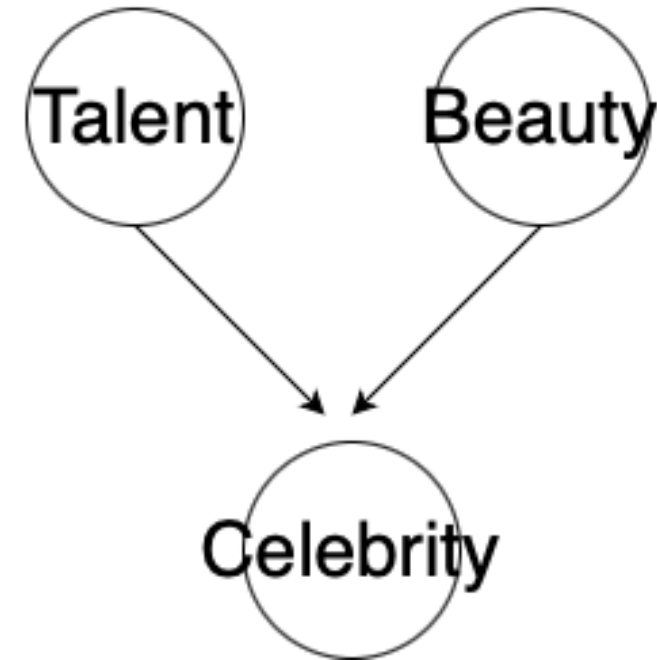


[8] Bayesian Net Representation (causal and effect examples)

[Examples of Bayes net that encodes Cause and Effect]



From the book “The book of Why” by Judea Peral

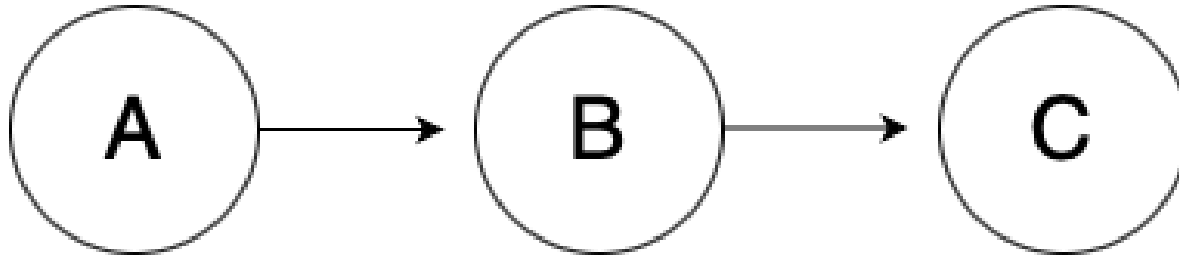


- we can design a Bayesian net based on prior knowledge about causal relationship. (assumption)

- Elementary Blocks of Bayesian Net (three vertexes and two edges)
 - chain structure
 - tent structure
 - V-structure

[1] Elementary Blocks (chain structure)

- **Chain Structure**: the middle node of chain breaks into two.



B mediator screen off information about A from C: $A \perp C \mid B$

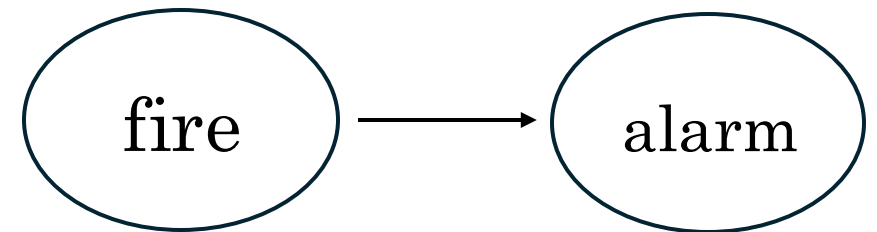
[2] Elementary Blocks (chain structure example)

- **Chain Structure:** the middle node of chain breaks into two.



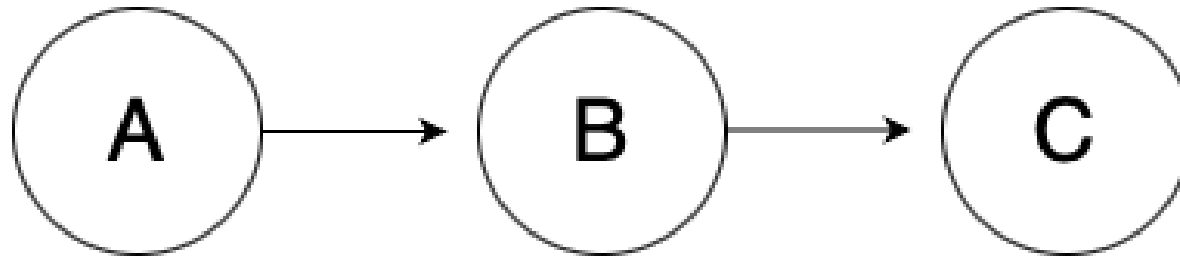
the fire by itself does not set off an alarm.

Q: is it possible? fire is a cause for the alarm?



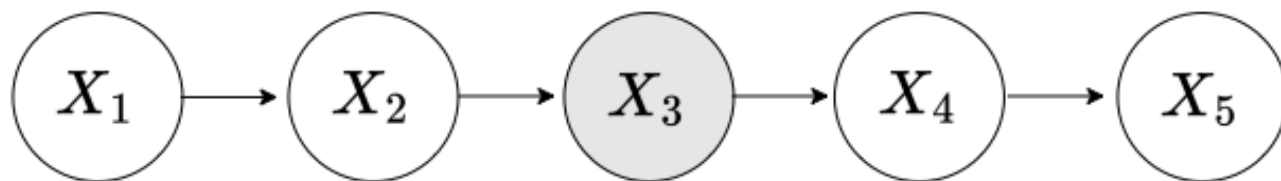
[3] Elementary Blocks (chain structure)

Q: a possible conclusion about causation from the chain structure ?



[4] Elementary Blocks (extended chain structure)

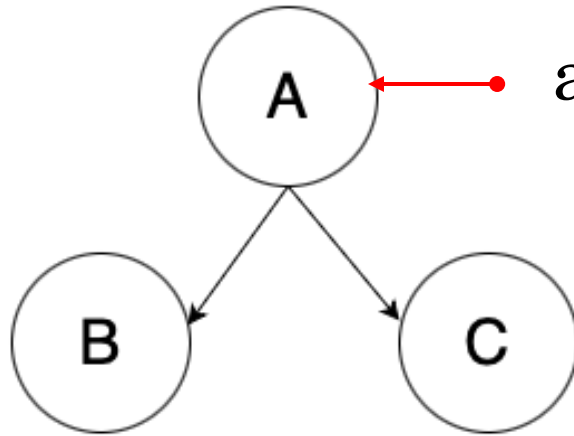
- extended chain structure: the middle node of chain breaks into two.



$$\begin{aligned} P(X_1, X_5 | X_3) &= \frac{P(X_1, X_3, X_5)}{P(X_3)} \\ &= \frac{\sum_{X_2, X_4} P(X_1, X_2, \dots, X_5)}{P(X_3)} \\ &= \frac{\sum_{X_2, X_4} P(X_1, X_2, X_3) P(X_4, X_5 | X_1, X_2, X_3)}{P(X_3)} \\ &= \sum_{X_2, X_4} P(X_1, X_2 | X_3) P(X_4, X_5 | X_3) \\ &= P(X_1 | X_3) P(X_5 | X_3) \end{aligned}$$

[5] Elementary Blocks (tent structure)

- tent (fork) structure: root node separates its children

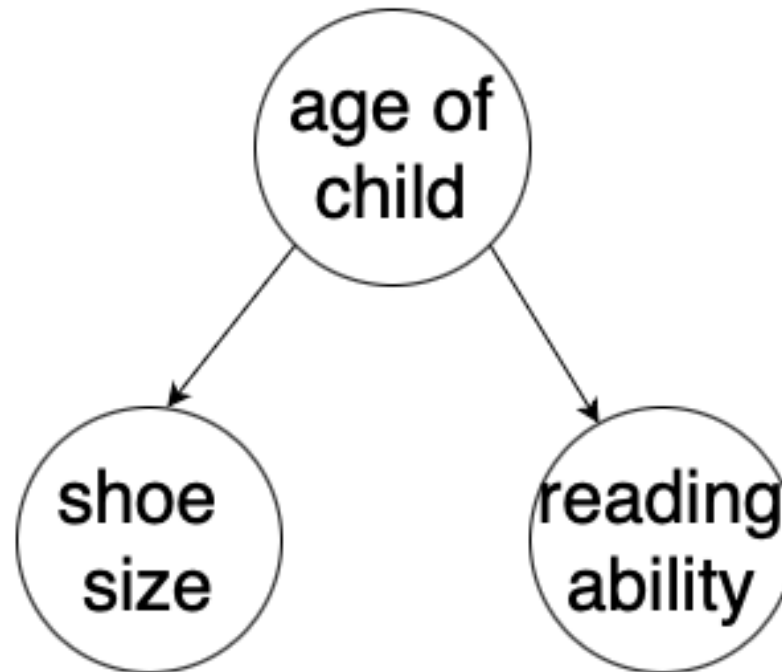


- the cofounder A makes B and C are statistically correlated.
- B and C are correlated without A
- $B \perp C \mid A$

$$\begin{aligned} P[B, C|A] &= \frac{P[ABC]}{P[A]} \\ &= \frac{P[AB]P[C|A]}{P[A]} \\ &= P[B|A] \cdot P[C|A] \end{aligned}$$

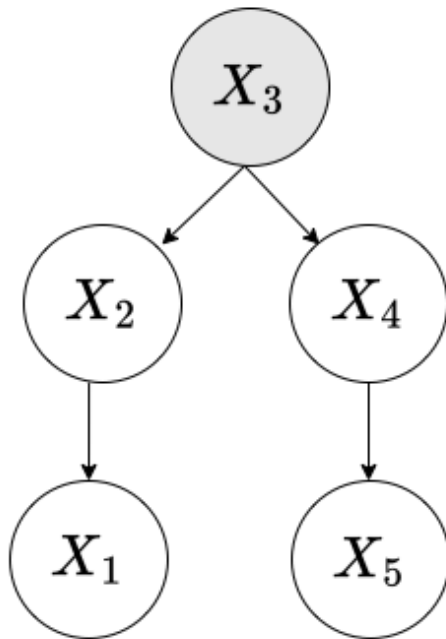
[6] Elementary Blocks (tent structure example)

- tent (Fork) structure: root node separates its children



[7] Elementary Blocks (extended tent structure example)

- extended tent structure: root node separates its children

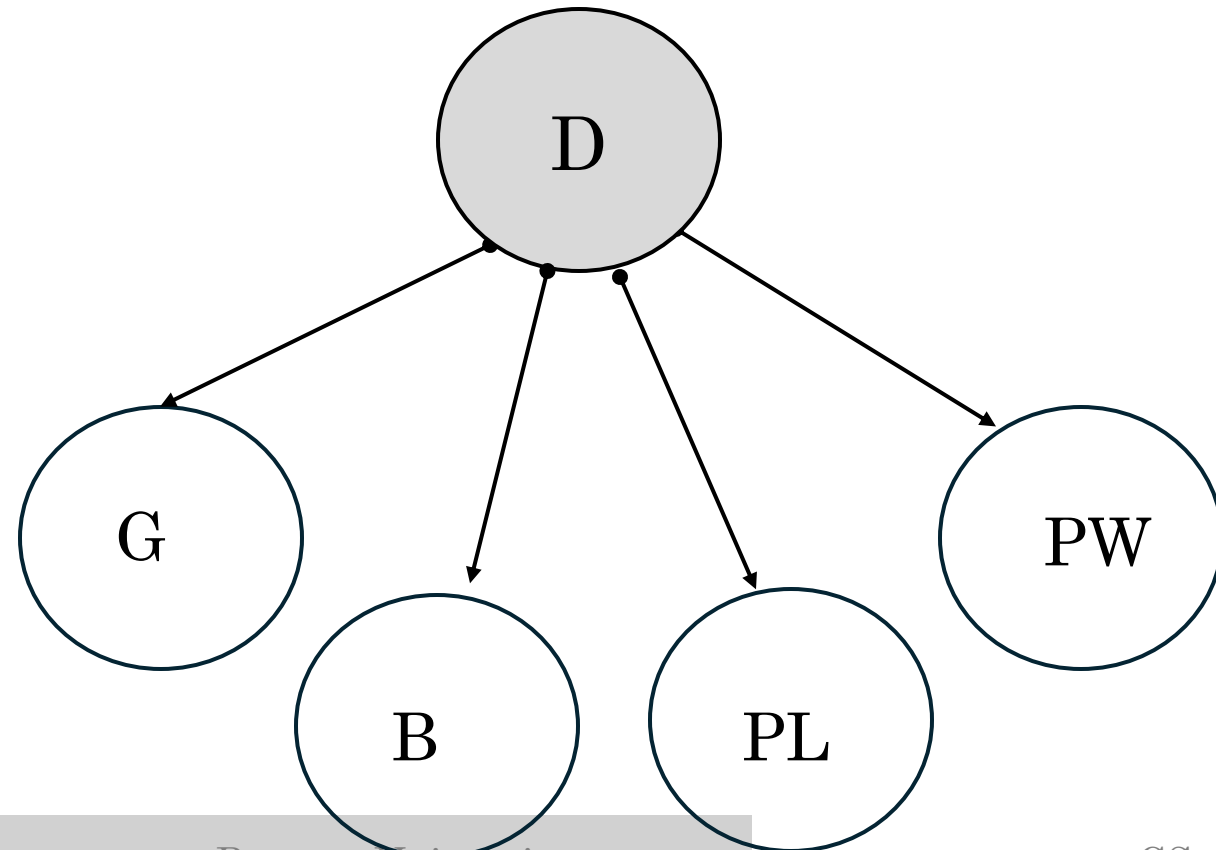


$$\begin{aligned} P(X_1, X_5 | X_3) &= \frac{P(X_1, X_3, X_5)}{P(X_3)} \\ &= \frac{\sum_{X_2, X_4} P(X_1, X_2, \dots, X_5)}{P(X_3)} \\ &= \frac{\sum_{X_2, X_4} P(X_1, X_2, X_3) P(X_4, X_5 | X_1, X_2, X_3)}{P(X_3)} \\ &= \sum_{X_2, X_4} P(X_1, X_2 | X_3) P(X_4, X_5 | X_3) \\ &= P(X_1 | X_3) P(X_5 | X_3) \end{aligned}$$

[8] Elementary Blocks (tent structure example)

[example of tent structure]

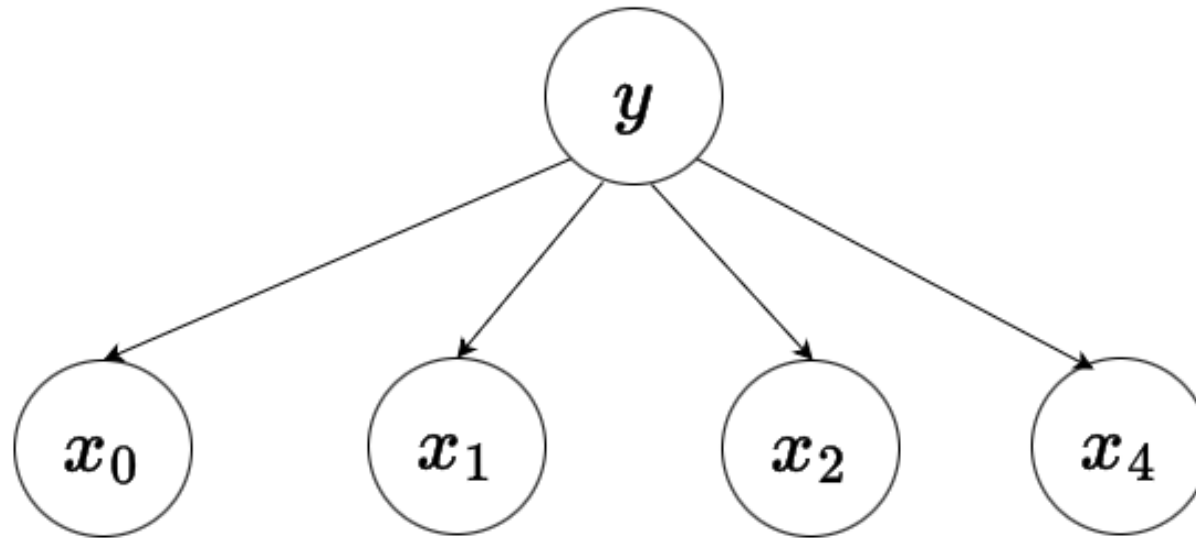
In the hw1, the feature of SL/ SW/ PL/ PW were conditionally independent given Iris information.



- when Iris information observed, SL/ SW/ PL/ PW becomes independent.

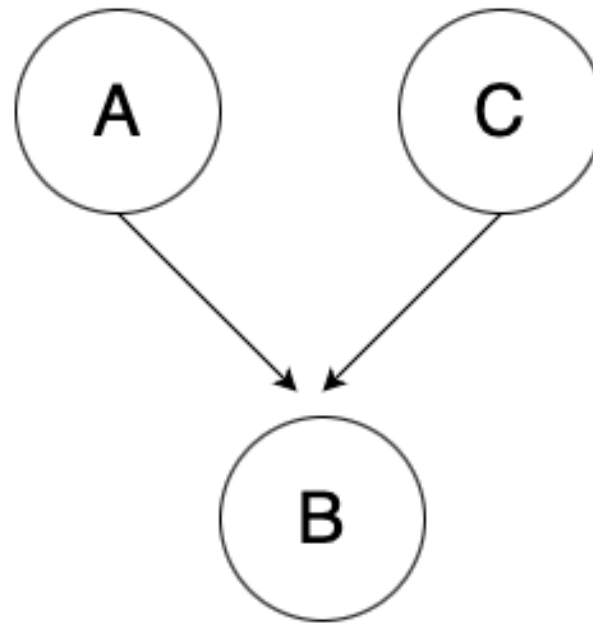
[9] Elementary Blocks (tent structure of naïve Bayes)

Conditional Independence Assumption for Naïve Bayes

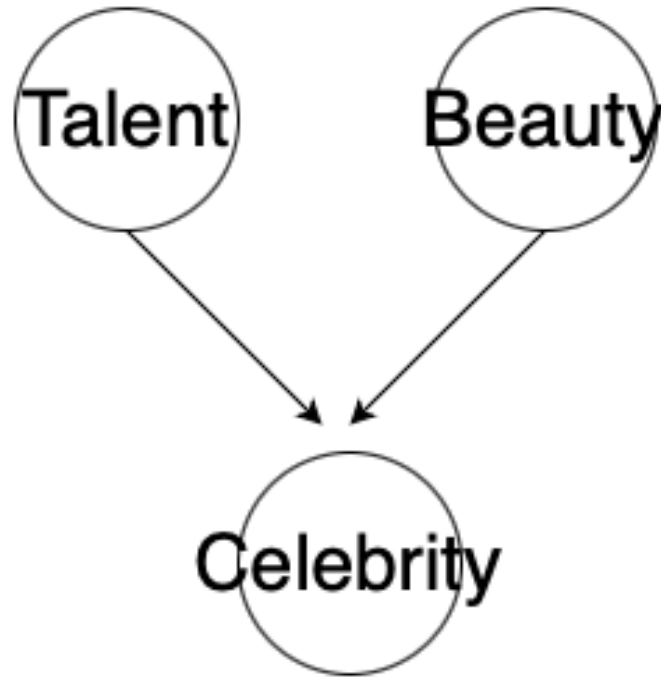


[10] Elementary Blocks (V structure)

- *V*-Structure (collider):
conditioning on a common child/ (or a descendant of the child) or at the bottom of a v-structure makes its parents become dependent.



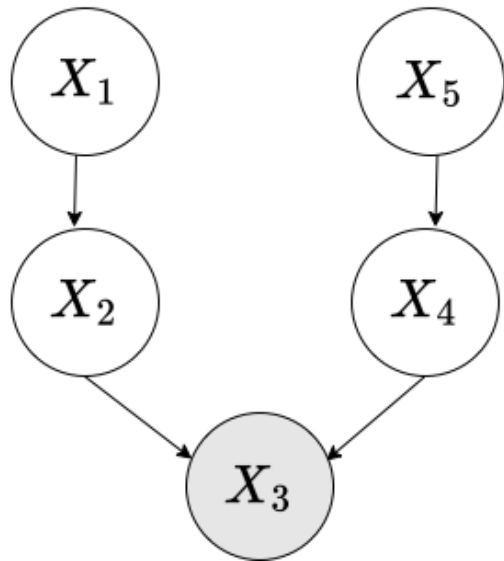
[11] Elementary Blocks (V structure example)



both talent and beauty contribute to an actor's success, but beauty and talent are completely uncorrelated. If a celebrity is a particularly good actor that explains away the person's success, the person does not need to be more beautiful than the average person.

[12] Elementary Blocks (extended V structure example)

- V-Structure (collider):
conditioning on a common child/ (or a descendant of the child) or at the bottom of a v-structure makes its parents become dependent.



$$\begin{aligned} P(X_1, X_5 | X_3) &= \frac{P(X_1, X_3, X_5)}{P(X_3)} \\ &= \frac{\sum_{X_2, X_4} P(X_1, X_2, \dots, X_5)}{P(X_3)} \\ &= \frac{\sum_{X_2, X_4} P(X_1, X_2) P(X_4, X_5) P(X_3 | X_2, X_4)}{P(X_3)} \\ &\neq P(X_1 | X_3) P(X_5 | X_3) \end{aligned}$$

- D-separation (a criterion to decide a set of A and B independent give e)
 $[A \perp B | e]$

[1] D separation (definition)

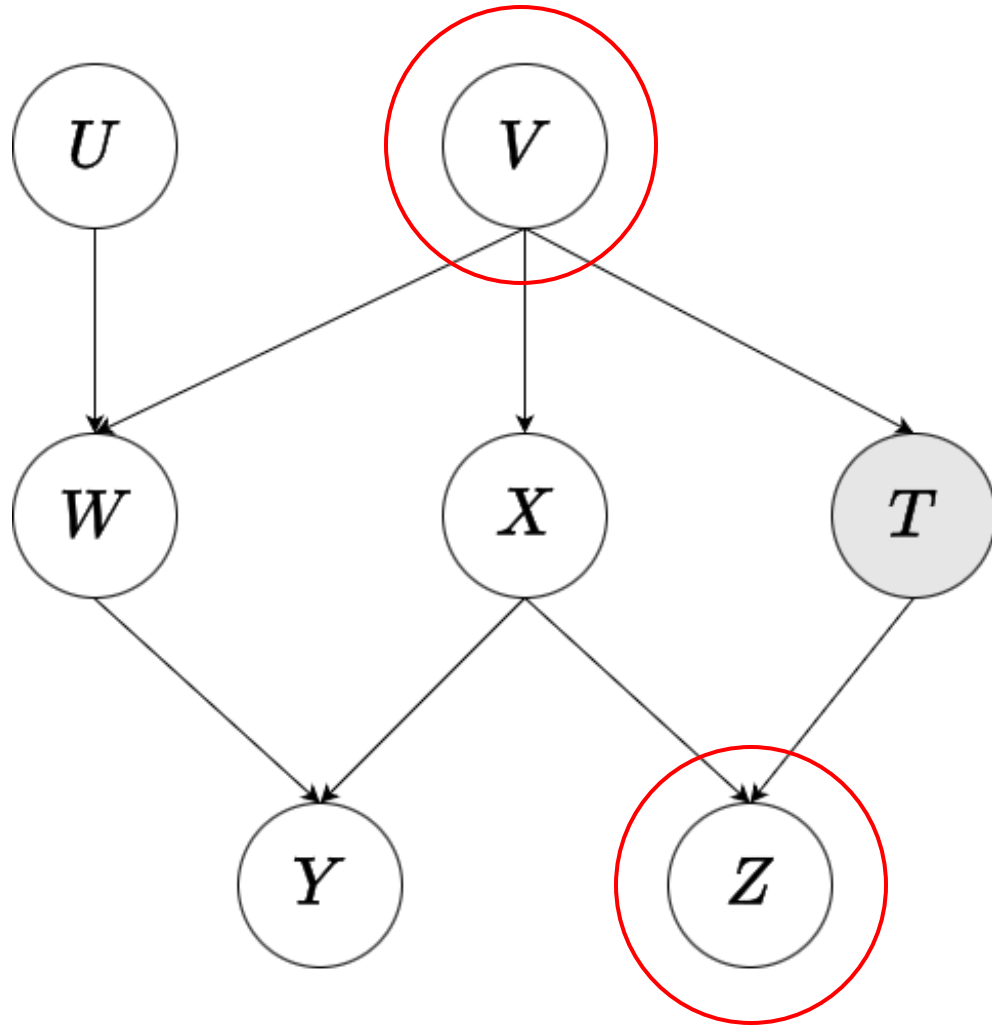
[D separation between two nodes]

The node A is **d-separated** from B by a set of observation e
 $\Leftrightarrow$ each (all) undirected path from A to B is d-separated.

$$\Leftrightarrow [A \perp B | e]$$

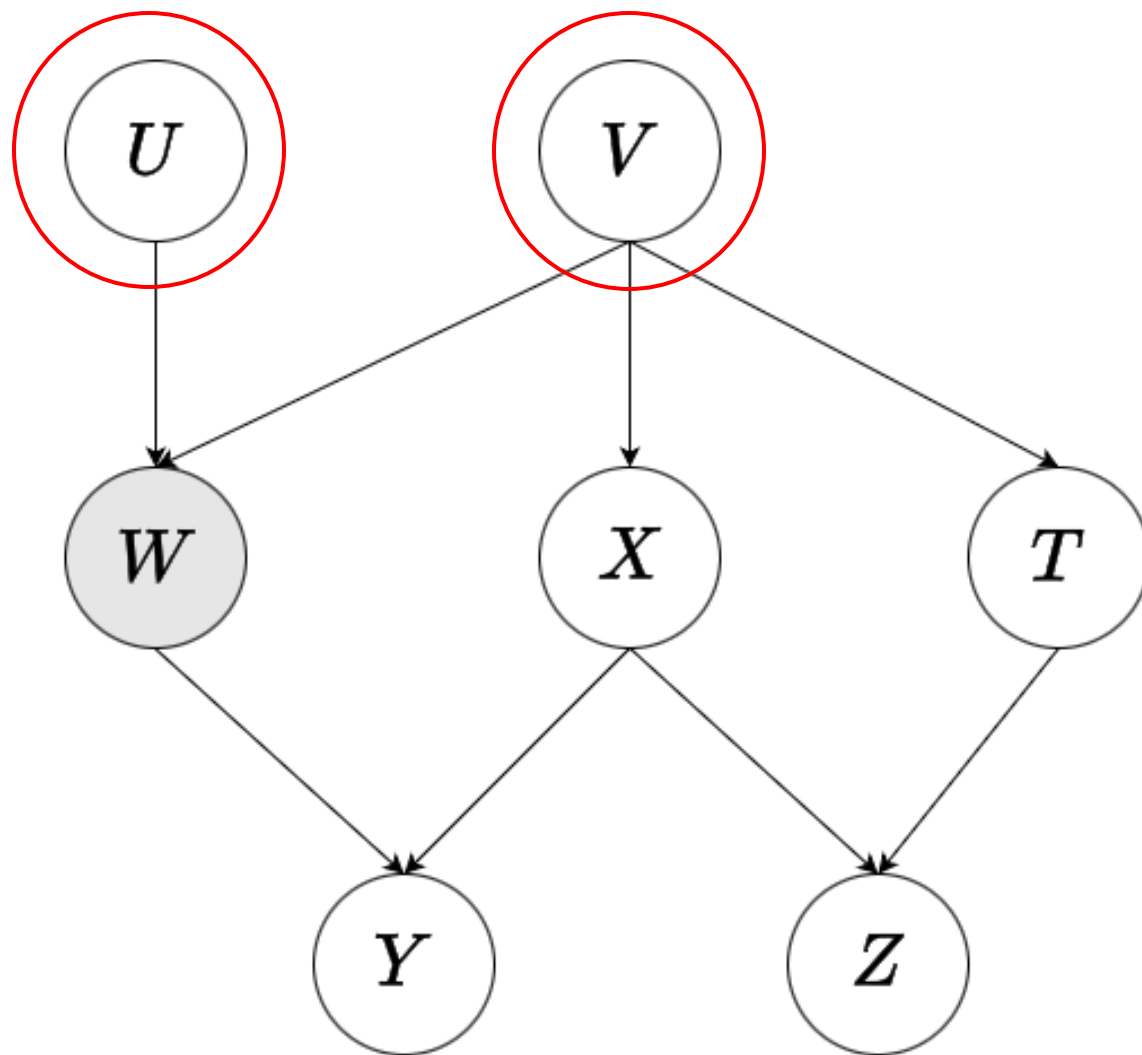
Q: What if we find a path between A and B that is not d-separated?

[2] D separation (example1)



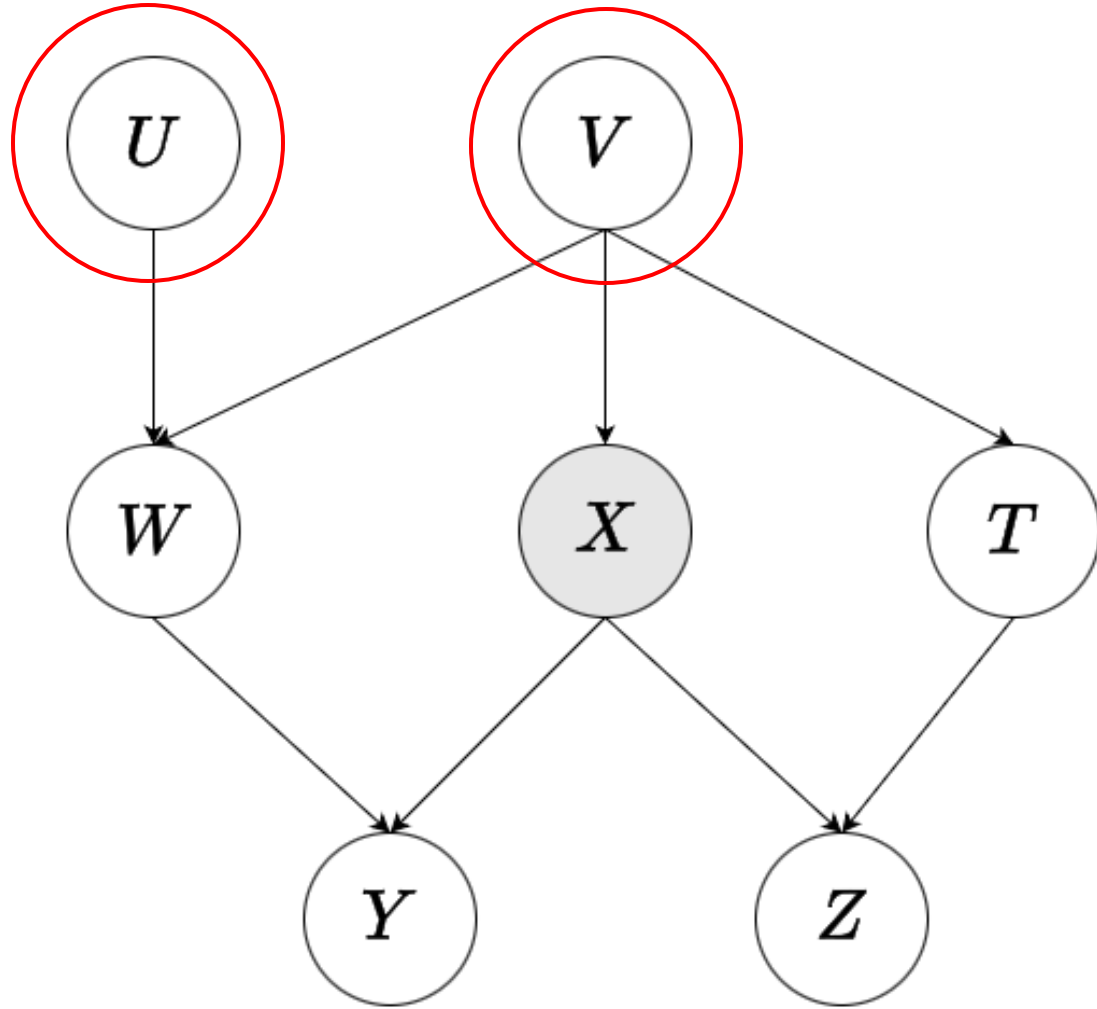
$Q: [V \perp Z|T]?$

[3] D separation (example2)



$Q: [U \perp V | W]?$

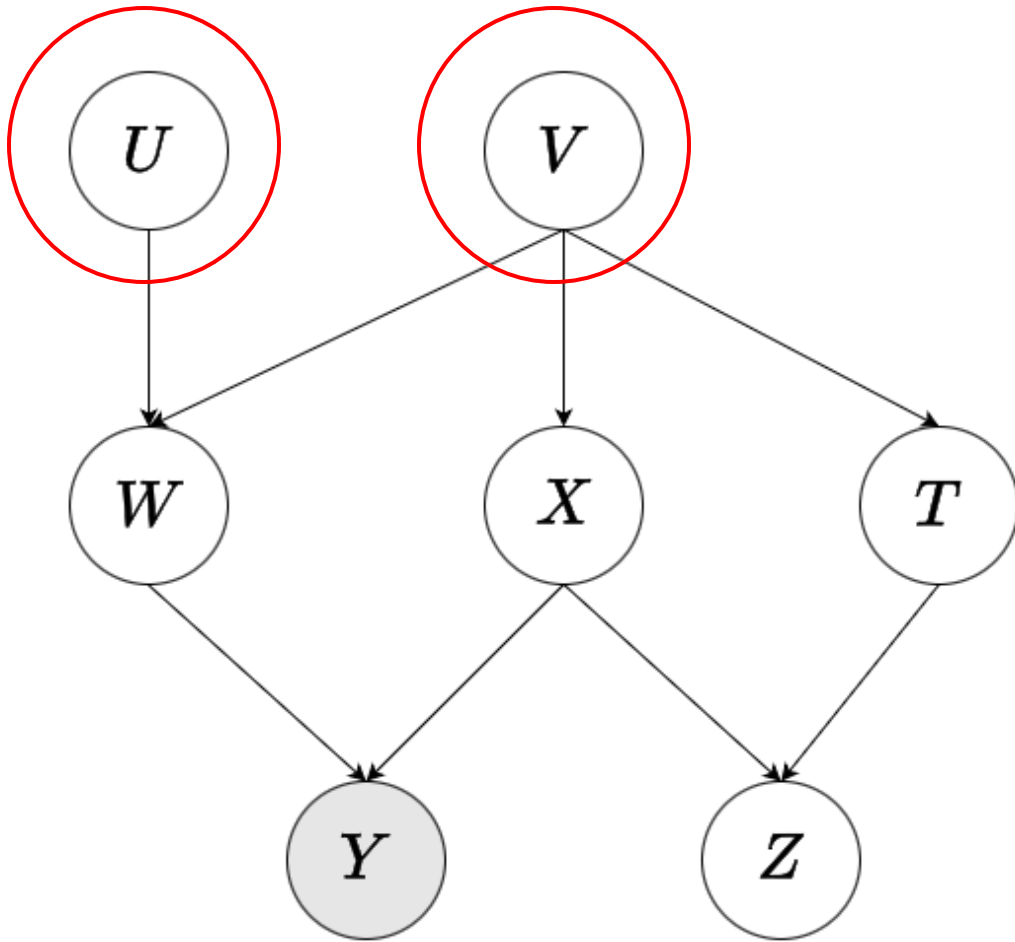
[4] D separation (example3)



$Q: [U \perp V | X]?$

[5] D separation (example4)

$Q: [U \perp V | Y]?$



[6] D separation (why it matters?)

How D-separation matter?

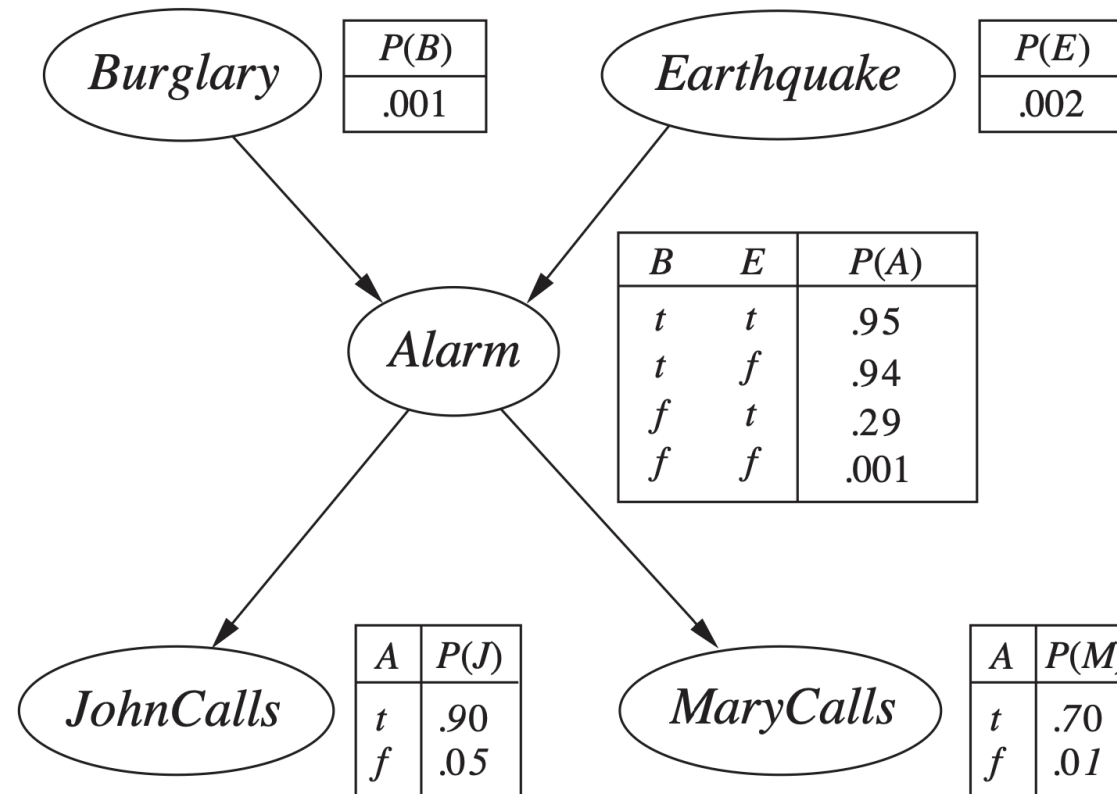
How $A \perp B \mid e$ matter?

What conclusion can be drive from it about causal relationship ?

[7] D separation (why it matters?)

In the previous slide,
Depending on the order of variable introduction,
the same density can be represented by various ways.

[8] D separation (why it matters?)

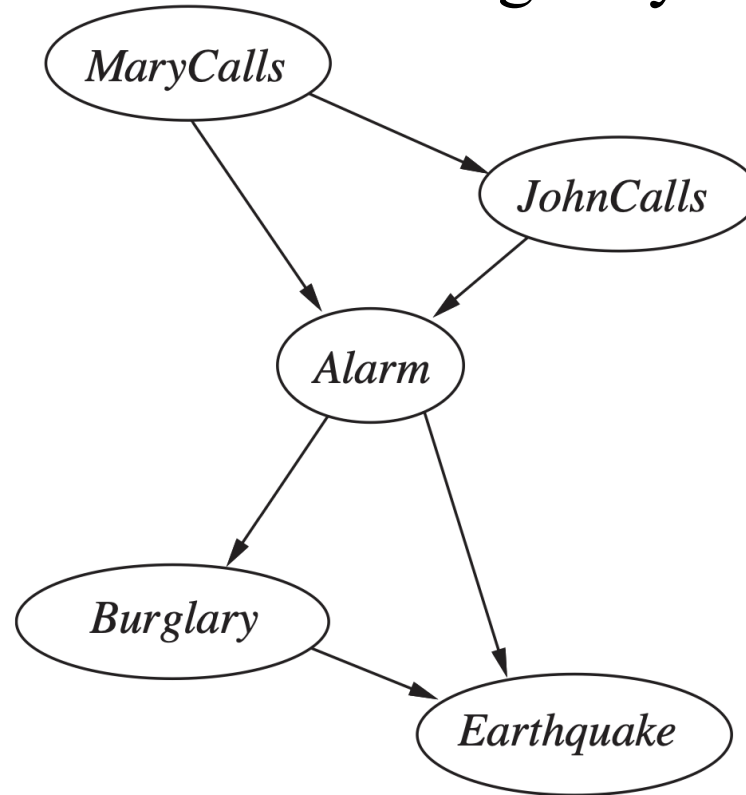


How the structure will be changed as the order of
MaryCalls → JohnCalls → Alarm → Burglary → Earthquake?

[9] D separation (why it matters?)

Bayesian network based on new order

:MaryCalls $\rightarrow$ JohnCalls $\rightarrow$ Alarm $\rightarrow$ Burglary $\rightarrow$ Earthquake?



From Figure 14.3 in Artificial Intelligence: A Modern Approach by Russell

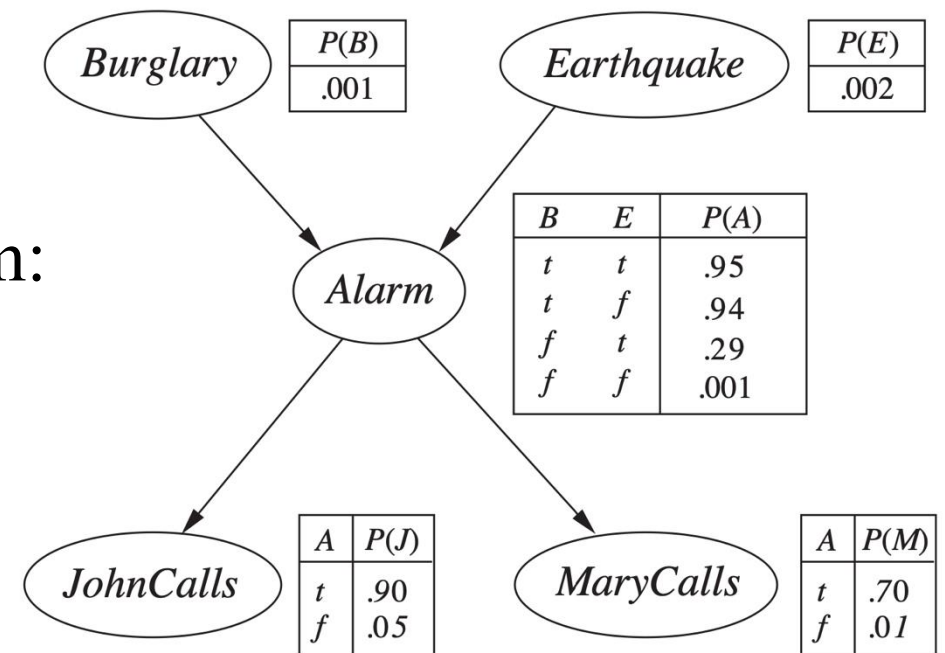
[10] D separation (why it matters?)

Bayesian network based on new order

:MaryCalls → JohnCalls → Alarm → Burglary → Earthquake?

For the new structure, the following conditional independence are derived from the original structure.

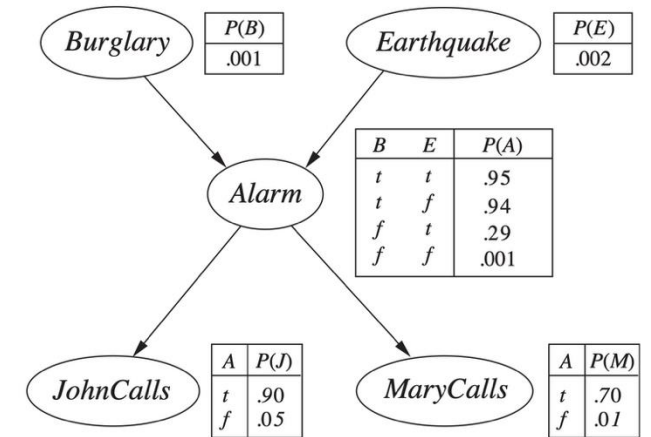
- Burglary $\perp$ MaryCalls and JohnCalls | Alarm:
- Earthquake $\perp$ MaryCalls, JohnCalls | Alarm
- Earthquake $\not\perp$ Burglary | Alarm



[11] D separation (why it matters?)

Bayesian network based on new order

:MaryCalls $\rightarrow$ JohnCalls $\rightarrow$ Alarm $\rightarrow$ Burglary $\rightarrow$ Earthquake?



- Inference: Computing the Posterior Probability
how to compute $P[\text{Burglary} \mid \text{MaryCalls} + \& \text{JohnCalls} +]?$

**handling the posterior queries is the main reason of Bayesian net is invented for.
(reduce the computational complexity in marginalization)

**Inference (computing posterior prob)

- without CI

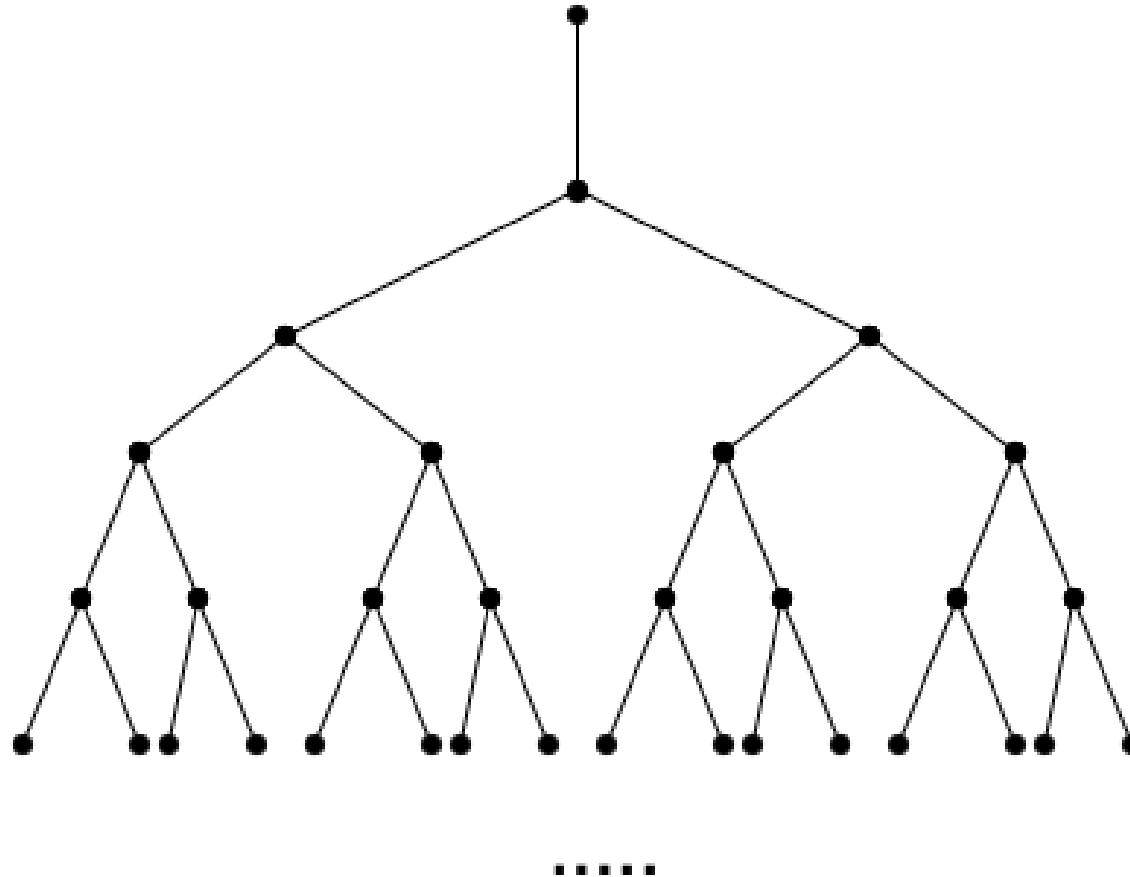
$$P(X_{36}) = \sum_{X_1} \sum_{X_2} \dots \sum_{X_{35}} \sum_{X_{37}} \dots \sum_{X_{1000}} P(X_1, X_2, \dots, X_{35}, X_{36}, X_{37}, \dots, X_{1000})$$

- with CI

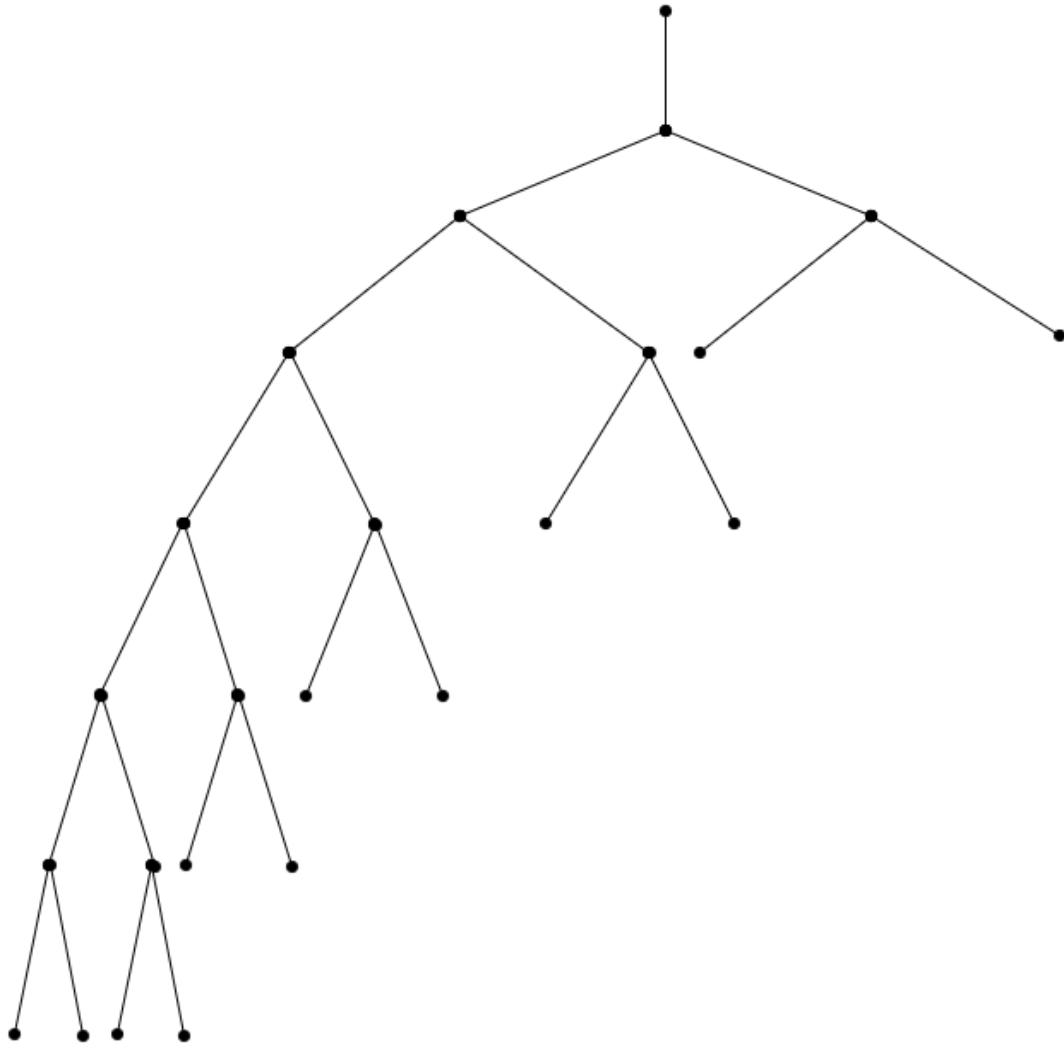
$$\begin{aligned} P(X_{36}) &= \sum_{X_1} \sum_{X_2} \dots \sum_{X_{35}} \sum_{X_{37}} \dots \sum_{X_{1000}} P(X_1)P(X_2|X_1), \dots, P(X_{36}|X_{35})P(X_{37}|X_{36}), \dots P(X_{1000}|X_{999}) \\ &= \sum_{X_1} P(X_1) \sum_{X_2} P(X_2|X_1) \dots \sum_{X_{35}} P(X_{35}|X_{34})P(X_{36}|X_{35}) \sum_{X_{37}} P(X_{37}|X_{36}) \dots \sum_{X_{1000}} P(X_{1000}|X_{999}) \end{aligned}$$

**Inference (complexity without CI: 2^N)

Conditional Independence in a Bayesian net
reduces the computational complexities in computing the query.



**Inference (complexity with CI: $O(N)$)



without CI: $\sim O(2^N)$ operations

with CI (Markov Chain Structure): $\sim O(N)$

[1] Inference (computing posterior prob)

The general format of query

$$P(X|e) = \frac{P(X, e)}{P(e)} = \frac{\sum_H P(X, H, e)}{P(e)} = \alpha \sum_H P(X, H, e)$$

- X query variable
- H non-query variable
- e evidence variable

Q: Bayesian net provides a joint density,
so any type of query can be answered?

[2] Inference (computing posterior prob)

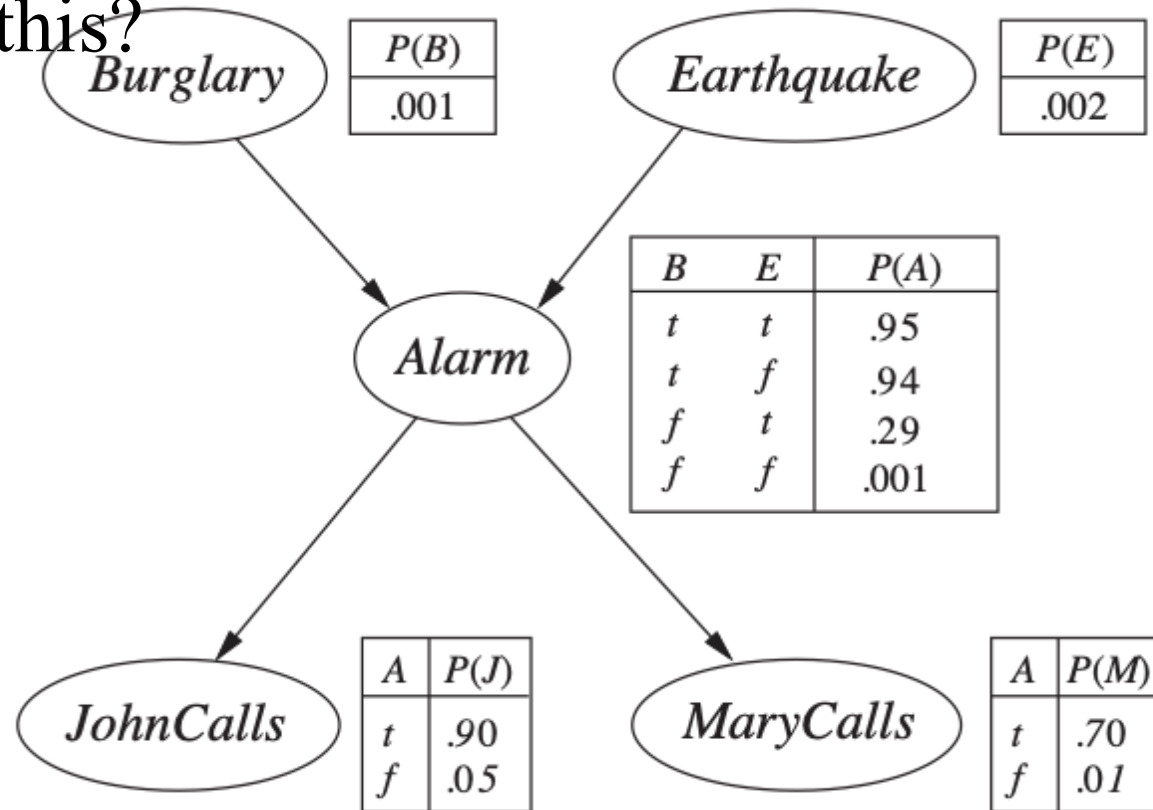
Given $P(X_1, X_2, X_3)$

$$P(X_1 | X_2 = x_2)? = \frac{P(X_1, X_2 = x_2)}{P(X_2 = x_2)} = \alpha \sum_{X_3} P(X_1, X_2 = x_2, X_3)?$$

[3] Inference (computing posterior prob example)

Q: $P[B = + \text{ and } M = +]$

how can we compute this?



[4] Inference (computing posterior prob example)

$$P[B|J = +, M = +] = \frac{P[B, J = +, M = +]}{P[J = +, M = +]} = \alpha \cdot P[B, J = +, M = +]$$
$$\alpha \cdot P[B, J = +, M = +] = \alpha \cdot \left[\begin{array}{l} P[B = +, J = +, M = +] \\ P[B = -, J = +, M = +] \end{array} \right]$$

Only we need to compute is $P[B, J = +, M = +]$.

$$\alpha (P[B = +, J = +, M = +] + P[B = -, J = +, M = +]) = 1$$

[5] Inference (computing posterior prob example)

How to compute

$$P[B, J = +, M = +] = \left| \frac{P[B = +, J = +, M = +]}{P[B = -, J = +, M = +]} \right|?$$

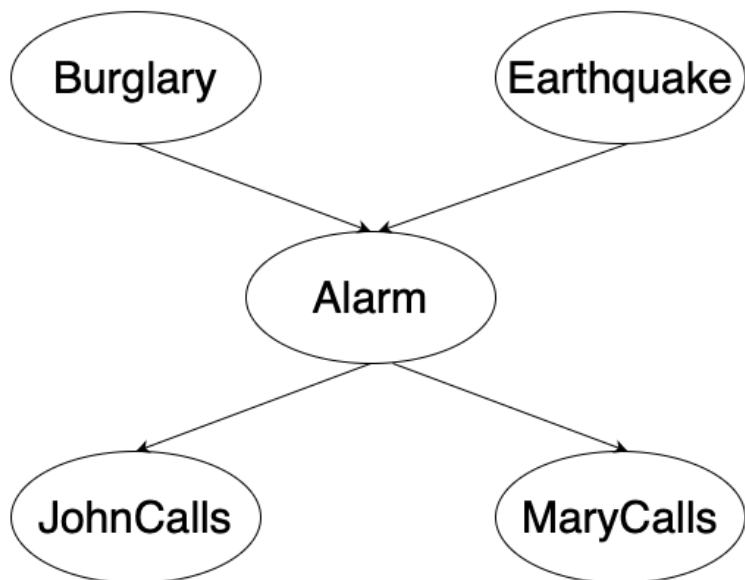
[6] Inference (computing posterior prob example)

By marginalization,

$$P[B, J = +, M = +] = \sum_E \sum_A P[B, E, A, J = +, M = +]$$

[7] Inference (computing posterior prob example)

$$\begin{aligned} P[B, J = +, M = +] &= \sum_E \sum_A P[B, E, A, J = +, M = +] \\ &= P[B] \sum_E P[E] \sum_A P[A|B, E] P[J = +|A] P[M = +|A] \\ &\quad \underbrace{f_1(B)} \quad \underbrace{f_2(E)} \quad \underbrace{f_3(A, B, E)} \quad \underbrace{f_4(A)} \quad \underbrace{f_5(A)} \end{aligned}$$

We need to compute from right to left direction.

[8] Inference (computing posterior prob example)

$$\begin{aligned} P[B, J = +, M = +] &= \sum_E \sum_A P[B, E, A, J = +, M = +] \\ &= P[B] \sum_E P[E] \sum_A \underbrace{P[A|B, E]}_{f_3(A, B, E)} \underbrace{P[J = +|A]}_{f_4(A)} \underbrace{P[M = +|A]}_{f_5(A)} \\ &\quad \underbrace{f_1(B) \quad f_2(E)}_{f_6(A)} \end{aligned}$$

$$f_6[A] = \begin{bmatrix} f_6[A = +] \\ f_6[A = -] \end{bmatrix} = \begin{bmatrix} P[J = +|A = +] \\ P[J = +|A = -] \end{bmatrix} \begin{bmatrix} P[M = +|A = +] \\ P[M = +|A = -] \end{bmatrix}$$

[9] Inference (computing posterior prob example)

$$\begin{aligned}
 P[B, J = +, M = +] &= \sum_E \sum_A P[B, E, A, J = +, M = +] \\
 &= \underbrace{P[B]}_{f_1(B)} \sum_E \underbrace{P[E]}_{f_2(E)} \sum_A \underbrace{P[A|B, E]}_{f_3(A, B, E)} \underbrace{P[J = +|A]}_{f_4(A)} \underbrace{P[M = +|A]}_{f_5(A)} \\
 &\quad \underbrace{\hspace{10em}}_{f_6(A)} \\
 &\quad \text{---} f_7(BE) = \sum_A f_3(A, B, E) f_6(A)
 \end{aligned}$$

$$\begin{aligned}
 f_7(BE) &= \sum_A P[A|BE] f_6[A] \\
 &= f_6(A+) \begin{bmatrix} P[A+|B+E+] & P[A+|B+E-] \\ P[A+|B-E+] & P[A+|B-E-] \end{bmatrix} + f_6(A-) \begin{bmatrix} P[A-|B+E+] & P[A-|B+E-] \\ P[A-|B-E+] & P[A-|B-E-] \end{bmatrix}
 \end{aligned}$$

[10] Inference (computing posterior prob example)

$$\begin{aligned} P[B, J = +, M = +] &= \sum_E \sum_A P[B, E, A, J = +, M = +] \\ &= \underbrace{P[B]}_{f_1(B)} \sum_E \underbrace{P[E]}_{f_2(E)} \sum_A \underbrace{P[A|B, E]}_{f_3(A, B, E)} \underbrace{P[J = +|A]}_{f_4(A)} \underbrace{P[M = +|A]}_{f_5(A)} \\ &\quad \underbrace{\hspace{10em}}_{f_7(BE)} \\ &\quad \underbrace{\hspace{10em}}_{f_8(B)} \end{aligned}$$
$$f_8(B) = \sum_E f_2(E) f_7(BE)$$

$$f_8(B) = P(E+)f_7(B, E+) + P(E-)f_7(B, E-)$$

[11] Inference (computing posterior prob example)

$$\begin{aligned}
 P[B, J = +, M = +] &= \sum_E \sum_A P[B, E, A, J = +, M = +] \\
 &= \underbrace{P[B]}_{f_1(B)} \sum_E \underbrace{P[E]}_{f_2(E)} \sum_A \underbrace{P[A|B, E]}_{f_3(A, B, E)} \underbrace{P[J = +|A]}_{f_4(A)} \underbrace{P[M = +|A]}_{f_5(A)} \\
 &\quad \underbrace{\left[\underbrace{f_2(E) f_3(A, B, E) f_4(A) f_5(A)}_{f_8(B) = \sum_E f_2(E) f_7(BE)} \right]}_{f_9(B) = P[B] f_8(B)} \\
 &\quad P[B, J = +, M = +] = f_1(B) f_7(B) \\
 &\quad P[B|J = +, M = +] = \alpha f_1(B) f_7(B) \\
 &\quad \alpha f_1(B = +) f_7(B = +) + \alpha f_1(B = -) f_7(B = -) = 1
 \end{aligned}$$